PID Controllers vs Fuzzy Logic Controllers in PMSM Motor

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Abstract: In a PMSM motor the conventional PID controllers show reduced performance of PMSM speed control under varying load parameters. This project aims to design a fuzzy logic controller and PID controller moreover comparing the Performance dynamic by comparing the Time Recovery and Disturbance Reject in the system. In addition, the system's robustness is examined by applying step changes in load torque at different time intervals, and the resulting speed dip and recovery characteristics are observed. The fuzzy controller is designed using linguistic variables and rule-based inference with triangular membership functions. Simulation results demonstrate that the fuzzy logic controller provides improved disturbance rejection, reduced speed deviation, and faster recovery compared to the conventional PID controller.

Keywords: Permanent Magnet Synchronous Motor (PMSM), PID Controller, Fuzzy Logic Controller (FLC), Speed Control, Load Disturbance, MATLAB/Simulink, Time Response Analysis, Disturbance Rejection.

I. INTRODUCTION

Permanent Magnet Synchronous Motors (PMSMs) are widely used in modern applications due to their high efficiency, high power, and fast dynamic response. These motors are commonly used in electric vehicles, robotics, and precision control systems where accurate speed regulation is essential. However, maintaining stable and accurate speed under varying load conditions remains a challenging task.

Conventional control techniques such as the Proportional-Integral-Derivative (PID) controller are widely used for motor speed control because of their simple structure and less complexity. Although PID controllers gives good performance of the system under steady state, their effectiveness decreases when the system is subjected to sudden load disturbances and parameter variations. This often results in increased settling time, larger overshoot, and poor disturbance rejection.

To overcome these limitations we aims to create a controller which works on rule based such as fuzzy logic base controller.
Fuzzy controllers do not require an accurate mathematical model of the system and are capable of handling nonlinearities and uncertainties effectively. By using linguistic rules and membership functions, fuzzy logic controllers can provide improved dynamic performance

In this paper, a comparative analysis of PID and fuzzy logic controllers for the speed control of a PMSM is presented. The system is modeled and simulated using **MATLAB/Simulink,** and the performance of both controllers is evaluated based on time response characteristics and disturbance rejection capability

II. PID CONTROLLER

The Proportional Integral Derivative (PID) controller is one of the most widely used controller because of its simple structure. It operates by continuously calculating the error between the reference speed and the actual motor speed, and applying a corrective control signal to minimize this error.

The control action of a PID controller is composed of three terms: proportional, integral, and derivative. The mathematical expression of the PID controller is given by:

$$u(t) = K_p\, e(t) + K_i \int e(t)\, dt + K_d\, (de(t)/dt)$$

where e(t) represents the error signal, defined as the difference between the reference speed and the actual speed of the motor. The proportional term provides an output that is directly proportional to the current error, the integral term eliminates steadystate error by correcting past errors, and the derivative term predicts future error based on its rate of change.

In the speed control of a Permanent Magnet Synchronous Motor , the PID controller generates the control torque required to the motor speed. Although PID controllers are effective under steady state conditions. Their performance is not so good when we add loads and vary them. This is one of the major draw back of the controller pid.

Despite these limitations, PID controllers are widely used due to their simplicity, and they serve as a baseline for comparison with advanced control techniques such as fuzzy logic control.

III. FUZZY LOGIC CONTROLLER

Fuzzy Logic Control (FLC) is an intelligent control technique that is widely used for systems where precise mathematical model is not the one and the system is non linear. Unlike conventional controllers, fuzzy logic controllers use linguistic variables and a rule-based approach to make control decisions, making them suitable for handling disturbances effectively.

In this work, a fuzzy logic controller is designed for the speed control of a Permanent Magnet Synchronous Motor . The controller takes two input variables: the speed error and the change in error. The error is defined as the difference between the reference speed and the actual motor speed, while the change in error represents the rate of variation of the error.

The output of the fuzzy controller is the control torque applied to the motor.

The design of the fuzzy controller involves three main stages: fuzzification, Rule based design interface, and defuzzification. In the fuzzification stage, the input variables are converted into fuzzy variables using membership functions. In this study, triangular membership functions are used, and the input and output variables are categorized into five linguistic terms: Negative Big (NB), Negative Small (NS), Zero (Z), Positive Small (PS), and Positive Big (PB).

Error \ ΔError	**NB**	**NS**	**Z**	**PS**	**PB**
NB	NB	NB	NS	NS	Z
NS	NB	NS	NS	Z	NS
Z	NS	NS	Z	PS	PS
PS	NS	Z	PS	PS	PB
PB	Z	PS	PS	PB	PB

In the defuzzification stage, the fuzzy output obtained from the inference process is converted back into a control signal.

IV. MATHEMATICAL MODELING

This section presents the mathematical formulation of the controllers and the Permanent Magnet Synchronous Motor (PMSM) used in this study.

A. PID Controller Tuning Equation

The control signal generated by a Proportional-Integral-Derivative (PID) controller is given by:

$u(t) = K_p\, e(t) + K_i \int e(t)\, dt + K_d\, (de(t)/dt)$

Taking Laplace transform:

$U(s) = (K_p + K_i/s + K_d\, s)\, E(s)$

where:
K_p = proportional gain
Ki = integral gain
K_d = derivative gain

B. Fuzzy Adaptive Tuning Equation

In the fuzzy logic controller, the control signal is generated based on scaled inputs using gain factors. The inputs are error (e) and change in error (Δe), and the output is control torque (u).

The fuzzy control law can be expressed as:
$u = K_u \cdot F(K_e\, e, K_{c_e}\, \Delta e)$

where:
K_e = input scaling factor for error
K_{c_e} = input scaling factor for change in error
K_u = output scaling factor
F(.) = nonlinear mapping defined by fuzzy inference system

This equation represents an adaptive control mechanism where the control action is determined by fuzzy rules instead of fixed mathematical relationships.

C. PMSM Mechanical Equation (Without Load)

The mechanical dynamics of the PMSM without load torque are given by:

$T_e = J\, (d\omega/dt) + B\omega$

Taking Laplace transform:

$T_e(s) = J\, s\, \omega(s) + B\, \omega(s)$

Rearranging:

$T_e(s) = (Js + B)\, \omega(s)$

Thus, the transfer function is:

$G(s) = \omega(s)/T_e(s) = 1 / (Js + B)$
Where we took J=0.01
And B=0.1

D. PMSM Mechanical Equation (With Load)

When load torque is considered, the equation becomes:

$T_e - T_L = J\, (d\omega/dt) + B\omega$

Taking Laplace transform:

$T_e(s) - T_L(s) = (Js + B)\, \omega(s)$

Rearranging:

$\omega(s) = [T_e(s) - T_L(s)] / (Js + B)$

Thus, the transfer function is:

$G(s) = \omega(s)/T_e(s) = 1 / (Js + B)$

and the load torque acts as a disturbance input:

$G_d(s) = \omega(s)/T_L(s) = -1 / (Js + B)$

The above mathematical models are used to design and simulate the PID and fuzzy logic controllers for PMSM speed control **in MATLAB/Simulink**.

V. SIMULATION RESULTS

1. PID Performance Without Load Varying

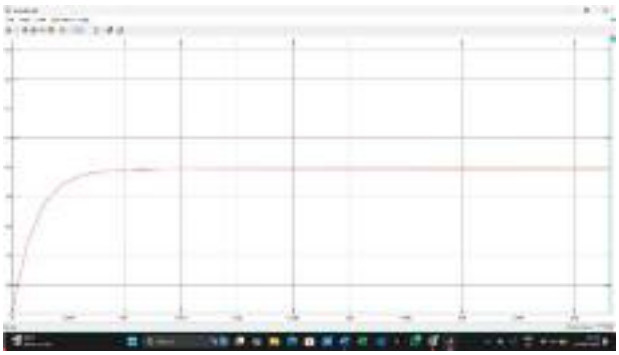
2. PID Performance With Load Varying

Disturbance Rejection Performance

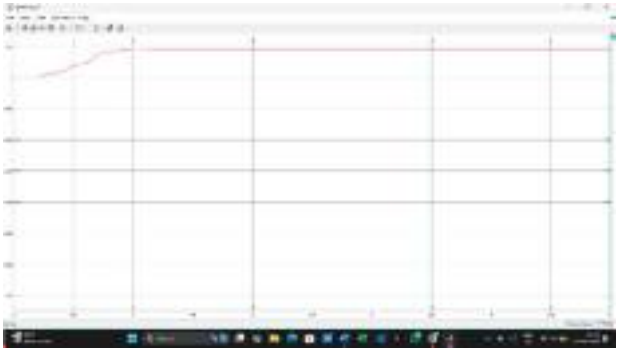
1. **Minimum Speed dip after Disturbance**
   Speed dip = 19 rad/s
2. **Recovary time after applying 3 loads at 3 different times -2% to +2% tolerance**
   T_R=> 1.25-0.19= 1.065s

3.FUZZY Rule Based .fis file making

4. FUZZY Performance Without Load Varying



5. FUZZY Performance With Load Varying



Disturbance Rejection Performance

1. **Minimum Speed dip after Disturbance**
   Speed dip =2 rad/s
2. **Recovary time after applying 3 loads at 3 different times -2% to +2% tolerance**
   T_R=> 0.0.43-0.041 = 0.002s

V. RESULTS AND DISCUSSION

The performance of the PID and Fuzzy Logic controllers for the speed control of the Permanent Magnet Synchronous Motor is evaluated using **MATLAB/Simulink** simulations. The comparison is carried out based on time response characteristics and disturbance rejection capability.

The Simulation Pictures shows the speed response of the PMSM for both PID and fuzzy logic controllers under no-load conditions and varying load conditions. It is observed that fuzzy gives better performance under varying load conditions as we can see in the simulation process.

To analyze the robustness of the controllers, load torque disturbances are applied at different time intervals, specifically 20 Nm at 0.2 s, 25 Nm at 0.4 s, and 30 Nm at 0.6s

The corresponding speed response is shown in the figures. It is evident that the PID controller experiences significant speed dips and requires more time to recover to the reference speed.

In contrast, the fuzzy logic controller shows smaller deviations in speed and achieves faster recovery after each disturbance.

The control torque generated by both controllers. The PID controller produces abrupt variations in control torque, whereas the fuzzy controller generates smoother control signals, contributing to better system stability.

Overall, the simulation results clearly demonstrate that the fuzzy logic controller provides superior performance in terms of faster dynamic response, reduced overshoot, improved disturbance rejection, and smoother control action. These characteristics make the fuzzy controller more suitable for

PMSM speed control applications under varying load conditions.

VII. CONCLUSION

In this paper, a comparative analysis of PID and Fuzzy Logic Controllers for the speed control of a Permanent Magnet Synchronous Motor (PMSM) has been presented. The system was modeled and simulated in **MATLAB/Simulink**, and the performance of both controllers was evaluated under different operating conditions.

The results obtained from the simulation demonstrate that the conventional PID controller provides satisfactory performance under steady-state conditions but shows limitations in handling load disturbances and dynamic changes. On the other hand, the fuzzy logic controller exhibits superior performance in terms of faster rise time, reduced settling time, minimal overshoot, and improved disturbance rejection capability.

Therefore, it can be concluded that fuzzy logic control is a more effective and reliable approach for PMSM speed control in applications involving nonlinearities and varying load conditions. Future work may include the implementation of hybrid or adaptive control techniques to further enhance system performance

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